

SUMMARY OF STUDENT RESOURCE II

NAME: AKINSANYA ABIMBOLA DEBORAH

STUDENT ID: VEPH/50B/AI0004

INTRODUCTION

From the student resource video, there were some data frames that were called into abbreviations. Here we called all the Python libraries to be used and like I mentioned earlier they included, pandas, numpy, matplotlib and seaborn. Some settings were also made to matplotlib and pandas.

- Import pandas as pd
- Import numpy as np
- Import matplotlib.pyplot as plt
- Import seaborn as sns.

Next the data was loaded using the 'pd.read_csv' function and we moved to the next step.

STEP 1: DATA UNDERSTANDING

- Dataframe
- Head and tail
- D-types
- Describe

Pandas by default don't display every column whenever a data is stored on it, rather it shows this sign of continuity (...).

Under the data frame, we have df.shape, df.head, df.columns, df.describe. df.columns is to list out all the columns in your data, df.describe () is for numeric data. This step gave more information about the characteristics of the dataset where we got to know the number of columns and rows and other details.

Note: There are string versions and numeric versions of a column in a data.

STEP 2: DATA PREPARATION

- Dropping irrelevant columns and rows.
- Identifying duplicated columns.
- Renaming columns.
- Feature creation.

However, there are functions used under this step;

To drop columns, we call `df.drop ()`.

To identify missing values, `df.isna (). sum ()`

For duplicates, `df duplicated`

On a subset column, `df.duplicated (subset =['column name'])`

For Query, `df.query ('column name == column name')`

The subset was then defined as `df` and `df.copy()` was added at the end so Python can understand that we are referencing the subset as a new data entirely and it doesn't relate to the original dataset.

STEP 3: FEATURE UNDERSTANDING

- Univariate analysis
- Plotting feature distribution
- Histogram
- KDE
- Boxplot.

Each feature (distribution and outliers in the data set).

To call the function for unique value:

For unique value. `df [ 'column name'], value_counts ()`.

Note: A single column in a dataframe is a series.

While writing the code for unique value, the teacher made use of the back slash sign `\`, this allows the break of your code into separate lines for a clean read, rather than a long line of code.

STEP 4: FEATURE RELATIONSHIP

- Scatter plot
- Heatmap correlation
- Pair plot
- Group by comparison

Moving on into advanced plots, we started using seaborn which is not as limited as matplotlib to view the relationship between two or more series. This gave insights into how such series relate and can be used to answer some questions. We also checked for correlation where the tutor mentioned that correlation can only be used on columns with numerical values.

STEP 5: ASK A QUESTION ABOUT THE DATA

- Try to answer a question you have about the data using a plot or statistics.

At this last step, we attempted a question which was “ What are the locations with the fastest roller coasters?”. We answered the question and also visualized it with a horizontal barchart.